

Oluwatobi Iyanuoluwa Akinmuleya

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PROFESSIONAL SUMMARY

Medical doctor focused on data-efficient artificial intelligence for medical imaging, with emphasis on resource-constrained healthcare settings. Research interests include self-supervised learning, multimodal representation learning, and nuclear medicine imaging

EDUCATION

Shenyang Medical College

Bachelor of Medicine, Bachelor of Surgery (MBBS)

Shenyang, China

Oct. 2018 – Jul. 2025

RESEARCH EXPERIENCE

Research Supervisor (Connexin Team)

Remote

CAMERA MRI Africa – Medical Imaging Bootcamp

Sep. 2025 – Jan. 2026

- Supervised participants within the Connexin team on PET imaging analysis workflows, providing technical guidance on medical imaging pipelines, dataset interpretation, and reproducible analysis practices.
- Facilitated collaborative research activities and contributed to building a structured learning environment emphasizing reproducibility, peer learning, and problem-solving in medical imaging.

SPARK (MICCAI-Endorsed Academy for African Medical Imaging Translation)

Remote

Research Fellow

Jan. 2026 – Present

- Developed and evaluated deep learning approaches for medical imaging research, including leading a low-field MRI representation learning research and benchmarking brain MRI foundation models using Nigerian patient data.
- Participated in the MICCAI Low-Field Pediatric Brain MRI Segmentation Challenge and contributed to reproducible AI pipelines for MRI preprocessing, model training, evaluation, and robust benchmarking.
- Collaborated with interdisciplinary teams across radiology, medical physics, and computer science to advance clinically relevant AI methods for equitable medical imaging applications.

PUBLICATIONS & PRESENTATIONS

Publications

- B. Gheiji, S. Moradi, M. Vatanparast, S. Shokrani, M.A. Bagherzadeh, D. Elyassirad, **O.I. Akinmuleya**, S. Goodarzi, M. Heidari-Foroozan, M. Moassefi, J.D. Rudie, E. Calabrese, R. Jain, S. Faghani. NeuroAIHub: An AI-Driven Framework for Automated Curation and Discovery of Neuroradiology Datasets. *American Journal of Neuroradiology*, Accepted Manuscript, 2026.
- Oluwatobi I. Akinmuleya**, P.F. Cohen, K. Kairemo. 68Ga-PSMA PET/CT/MRI in the initial diagnosis and staging of prostate cancer: A review. *Advances in Radiotherapy & Nuclear Medicine*, Volume 2(4), 2024, 4590.

Presentations

- The Role of 68Ga-PSMA PET/CT and PET/MRI in the Diagnosis of Primary Prostate Cancer and TNM Staging**, Poster Presentation, International Conference of Hybrid Imaging, International Atomic Energy Agency (IAEA), Vienna (2024).
- NeuroAI: An AI-Driven Framework for Automated Curation and Discovery of Neuroradiology Datasets**, Oral/Poster Presentation, American Society of Neuroradiology (ASNR) Annual Meeting, Philadelphia, USA (2025).
- Training 105 African Clinicians and Students in Open Neuroimaging Analysis: CAMERA's Dimensional Research Program**, Poster Presentation, International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting (2026).
- Quantifying Amyloid Burden in Alzheimer's Disease: A Reproducible Centiloid Pipeline for Analysis in Resource-Constrained Settings**, Poster Presentation, Alzheimer's Association International Conference (AAIC) (2026), Abstract ID: #4997.

SELECTED PROJECTS

Quantifying Amyloid Burden Across the Alzheimer’s Continuum: A Reproducible Centiloid Pipeline

Developed a reproducible neuroimaging pipeline for amyloid PET quantification using the Centiloid framework for Alzheimer’s disease analysis in resource-constrained settings. Implemented SUVR and Centiloid computation with validation against published standards, along with scalable batch processing and automated quality control workflows.

In-Hospital Mortality Prediction System | *Python, MLflow, Docker, GCP, Streamlit, Evidently AI, Grafana*

Built an end-to-end machine learning pipeline for in-hospital mortality prediction to support clinical decision-making in critical care settings. Designed and deployed an MLOps system using MLflow, Docker, and GCP with integrated monitoring via Evidently AI and Grafana dashboards.

Computer Vision Systems for Medical and Surveillance Imaging | *PyTorch, YOLOv11, OpenCV, MTCNN*

Built real-time face recognition and tracking system using MTCNN and InceptionResNet for identity verification in video streams. Developed and fine-tuned YOLOv11 models for object detection in surveillance and gastrointestinal polyp detection for clinical imaging applications.

VOLUNTEER EXPERIENCE

Volunteer Tutor | *Shenyang Medical College, China*

2019 – 2023

Provided academic support and tutoring to medical students in core preclinical and clinical subjects, facilitating peer-to-peer learning and reinforcing foundational medical knowledge.

Volunteer Ambassador | *Women in Data Science (WiDS) Worldwide*

2024 – Present

Represented and promoted WiDS initiatives focused on increasing participation of women in data science and AI, supporting outreach and community engagement activities to improve inclusivity in data science education.

Tutorial Reviewer | *IndabaX Ghana (Ghana Data Science Summit)*

2025 – Present

Reviewed tutorial submissions for technical accuracy, clarity, and educational value, providing structured feedback to improve content quality and accessibility of data science education across diverse learner levels.

HONORS & AWARDS

- **Best Overall Medical Student**, Shenyang Medical College (2019–2021).
- **Academic Excellence Scholarship**, Shenyang Medical College (2019–2024).

SELECTED WORK EXPERIENCE

AI Engineer (Internship)

May 2025 – Jul. 2025

Mnemos

- Developed AI-driven solutions to support medical education, contributing to tools for creating and enhancing learning resources for medical students across Africa.
- Applied machine learning and natural language processing approaches to improve educational content workflows, collaborating with product and engineering teams to translate AI concepts into user-facing features.

Subinternship

Feb. 2023 – May 2024

Greater Accra Regional Hospital

Accra, Ghana

- Completed clinical subinternship training across multiple hospital departments, gaining hands-on experience in patient management and diagnostic workflows.
- Assisted in clinical case evaluation and supported decision-making processes under supervision of senior clinicians.
- Developed foundational clinical competence in hospital-based healthcare delivery within a resource-constrained setting.

SKILLS

Programming Languages: Python, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, MONAI

Medical Imaging & Scientific Computing: nibabel, SimpleITK, OpenCV, torchvision

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Web & API Development: Django, Flask, FastAPI, Streamlit, React, TailwindCSS, HTML/CSS, RESTful APIs

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle